

HDOC - Day 13

Even and Odd Digit Operations

Algorithm

1. Start. 2. Read positive integer n. 3. Display menu. 4. For option 1, extract every digit; if digit%2==0 increment even count, otherwise increment odd count. 5. For option 2, extract every digit and add it to sumEven or sumOdd according to parity. 6. Display results. 7. Repeat until Exit. 8. Stop.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	12045; choice 1	Even=3, Odd=2	Even=3, Odd=2	Pass
2	12345; choice 2	Sum_even=6, Sum_odd=9	6, 9	Pass
3	2468; choice 1	Even=4, Odd=0	Even=4, Odd=0	Pass

C Program

```
#include <stdio.h>
int main() {
    int n, choice, temp, d, even, odd, sumEven, sumOdd;
    do { printf("Enter a positive integer: "); scanf("%d", &n); } while(n <= 0);
    do {
        printf("\n1. Count even and odd digits\n2. Sum even and odd digits\n3. Exit\nEnter choice: ");
        scanf("%d", &choice);
        switch(choice) {
            case 1:
                temp = n; even = odd = 0;
                while(temp > 0) {
                    d = temp % 10;
                    if(d % 2 == 0) even++; else odd++;
                    temp /= 10;
                }
                printf("Even_dig = %d\nOdd_dig = %d\n", even, odd); break;
            case 2:
                temp = n; sumEven = sumOdd = 0;
                while(temp > 0) {
                    d = temp % 10;
                    if(d % 2 == 0) sumEven += d; else sumOdd += d;
                    temp /= 10;
                }
                printf("Sum_even = %d\nSum_odd = %d\n", sumEven, sumOdd); break;
            case 3: break;
            default: printf("Invalid choice.\n");
        }
    } while(choice != 3);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day13.c -o day13
puppie@Anvikas-MacBook-Air Desktop % ./day13
Enter a positive integer: 12045
Enter choice: 1
Even_dig = 3
Odd_dig = 2
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day13
Enter a positive integer: 12345
Enter choice: 2
Sum_even = 6
Sum_odd = 9
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day13
Enter a positive integer: 2468
Enter choice: 1
Even_dig = 4
Odd_dig = 0
```

Result: The menu-driven program was compiled, executed and tested successfully against the prepared test cases.